1. TITLE OF THE INVENTION

A SYSTEM AND METHOD FOR ASSESSING GROUNDWATER CONTAMINATION–INDUCED LONG-TERM CANCER RISK USING DEPTH-RESOLVED CUMULATIVE EXPOSURE MODELING

2. APPLICANT

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Address: *(To be filled as per Form-1)*

3. FIELD OF THE INVENTION

The present invention relates to groundwater contamination assessment and environmental health risk analysis. More particularly, the invention relates to a **computer-implemented system and method** for assessing **long-term cancer risk associated with chronic exposure to contaminated groundwater**, using depth-resolved cumulative exposure modeling.

4. BACKGROUND OF THE INVENTION

Groundwater is a primary source of drinking water in rural regions of India. In many such regions, long-term exposure to groundwater contaminated with arsenic, fluoride, nitrates, heavy metals, and pesticide residues has been associated with an increased incidence of cancer and other chronic diseases.

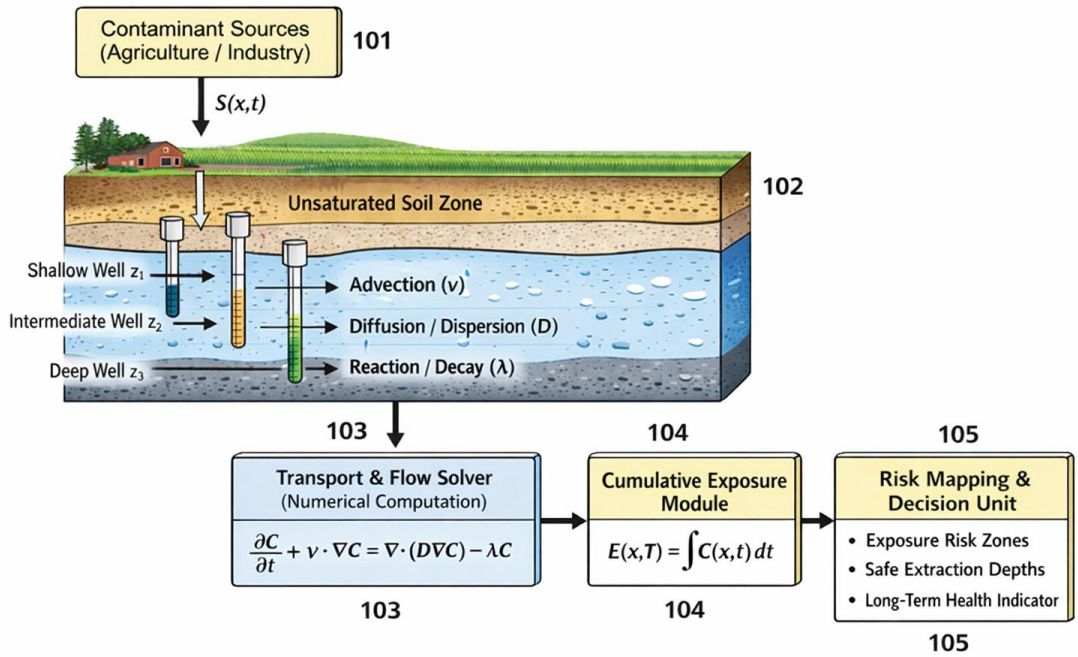
Cancer arising from environmental exposure typically results from **long-term, low-dose cumulative exposure**, rather than short-term concentration peaks. However, existing groundwater assessment approaches rely largely on isolated sampling or instantaneous concentration limits, which do not adequately represent cumulative exposure over extended periods.

Further, groundwater extraction occurs at varying depths, and contaminant transport and accumulation may vary significantly across these depths. Existing methods do not sufficiently account for **depth-wise exposure differences**.

The present invention is motivated by **field observations of clustered cancer incidence in groundwater-dependent rural populations**, where systematic depth-resolved

exposure assessment is lacking. There is therefore a need for a **quantitative, mechanistic, and non-clinical framework** to assess groundwater contamination-induced cancer risk.

FIG. 1



5. OBJECTS OF THE INVENTION

The primary objects of the present invention are:

1. To provide a system for modeling contaminant transport in groundwater.
2. To assess depth-wise contaminant accumulation in aquifers.
3. To compute cumulative exposure relevant to long-term cancer risk.
4. To identify groundwater extraction depths associated with reduced cancer risk.
5. To support preventive groundwater management and public health planning.

6. SUMMARY OF THE INVENTION

The present invention provides a **computer-implemented system and method** for assessing groundwater contamination–induced long-term cancer risk.

The invention models contaminant transport in groundwater using advection, diffusion, and reaction mechanisms. Contaminant concentrations are computed across multiple groundwater extraction depths. A cumulative exposure metric is calculated by integrating contaminant concentration over time, which serves as a **risk proxy for long-term cancer development**.

The system generates depth-resolved exposure profiles and risk indicators to support groundwater management decisions aimed at reducing environmentally induced cancer risk. The invention does not perform medical diagnosis and does not rely on individual clinical data.

7. BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a system-level schematic of groundwater contamination transport, cumulative exposure computation, and cancer risk assessment in accordance with an embodiment of the present invention.

8. DETAILED DESCRIPTION OF THE INVENTION

8.1 System Overview

Referring to **FIG. 1**, the system comprises the following components:

(a) Contaminant Source Module (101)

Configured to represent time-dependent contaminant inputs originating from agricultural, industrial, or natural sources, expressed as a source term $S(x, t)$.

(b) Subsurface Transport Domain (102)

Represents unsaturated soil and saturated aquifer regions as porous media. The domain resolves contaminant transport across multiple groundwater extraction depths.

(c) Transport and Flow Solver Module (103)

Configured to compute contaminant concentration $C(x, t)$ using an advection–diffusion–reaction equation:

$$\frac{\partial C}{\partial t} + \mathbf{v} \cdot \nabla C = \nabla \cdot (D \nabla C) - \lambda C + S(x,t)$$

Groundwater velocity is determined using Darcy's law. Numerical methods such as finite element or finite volume techniques are employed.

(d) Cumulative Exposure Module (104)

Configured to compute cumulative exposure:

$$E(x,T) = \int_0^T C(x,t)\, dt$$

The cumulative exposure represents long-term contaminant intake relevant to cancer risk.

(e) Risk Mapping and Decision Unit (105)

Configured to generate exposure-based risk zones, identify safer groundwater extraction depths, and provide decision-support outputs for cancer risk mitigation.

9. ADVANTAGES OF THE INVENTION

- Captures **long-term cumulative exposure**, unlike snapshot-based methods.
- Resolves **depth-dependent groundwater contamination behaviour**.
- Does not require clinical or personal health data.
- Supports preventive strategies for reducing cancer risk.
- Suitable for rural and data-scarce regions.

10. INDUSTRIAL APPLICABILITY

The invention is applicable in groundwater quality monitoring, environmental cancer risk assessment, rural water management, public health planning, and environmental regulatory systems.

11. CLAIMS

Claim 1 (Independent Claim)

A computer-implemented system for assessing groundwater contamination-induced long-term cancer risk, comprising:
a contaminant source module configured to define time-dependent contaminant inputs;
a subsurface transport domain representing a porous groundwater aquifer with depth-resolved extraction points;
a transport and flow solver configured to compute contaminant concentration using advection–diffusion–reaction equations;
a cumulative exposure module configured to compute time-integrated contaminant exposure; and
a risk mapping and decision unit configured to identify exposure risk zones and safer groundwater extraction depths.

Claim 2

The system as claimed in claim 1, wherein cumulative exposure is used as a proxy for long-term cancer risk assessment.

Claim 3

The system as claimed in claim 1, wherein the transport solver employs finite element or finite volume numerical methods.

Claim 4

The system as claimed in claim 1, wherein the system is informed by field-observed groundwater usage and contamination patterns.

12. ABSTRACT

A system and method for assessing groundwater contamination–induced long-term cancer risk are disclosed. The invention models contaminant transport in groundwater and computes cumulative exposure as a time-integrated concentration measure. Depth-resolved exposure profiles and risk indicators are generated to support preventive groundwater management and reduce environmentally induced cancer risk.